

$$y_j = \exp(u_j)$$

$$p_i = \frac{\partial \log G}{\partial u_i}$$

Now

$$G = \sum_{j=1}^J y_j$$

$$= \sum_{j=1}^J e^{u_j}$$

$$\log G = \log \sum_{j=1}^J e^{u_j}$$

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Chain rule:

$$\frac{\partial \log G}{\partial u_i} = \frac{1}{G} \frac{\partial G}{\partial u_i}$$

$$\text{Now } \frac{\partial G}{\partial u_i} = \frac{\partial \sum_{j=1}^J e^{u_j}}{\partial u_i}$$

$$= \frac{\partial (e^{u_1} + e^{u_2} + \dots + e^{u_J})}{\partial u_i}$$

$$= e^{u_i} \quad \text{(each term is constant to } u_i \text{)}$$

$$\Rightarrow p_i = \frac{e^{u_i}}{\sum_{j=1}^J e^{u_j}} \quad \text{Multinomial logit}$$

Conditions :

$$Q = \sum_{j=1}^J y_j$$

for $y_j \geq 0$ $\sum_{j=1}^J y_j \geq 0 \Rightarrow Q \geq 0$

$$Q(+y_1, \dots, +y_J) = \sum_{j=1}^J +y_j = + \sum_{j=1}^J y_j = +Q(y)$$

$\Rightarrow$ Homogenous of degree 1 here
Similarly we can do for degree k
with function $\sum_{j=1}^J y_j^k$

for limiting $y_j \rightarrow 0$

$$Q = \sum_{j=1}^J y_j = y_j + \sum_{i=1}^{j-1} y_i + \sum_{j+1}^J y_j = \infty$$

Now

$$(-1)^{s+1} \frac{\partial^s Q}{\partial y_{i_1} \dots \partial y_{i_s}}$$

for $s=1$

$$\frac{\partial Q}{\partial y_i} = \frac{\partial (y_1 + y_2 + \dots + y_i + \dots + y_J)}{\partial y_i} = 0 + 0 + \dots + 1 + \dots + 0$$

$$= 1$$

$$\Rightarrow (-1)^{1+1} 1 = 1 \geq 0$$

for $s=2$

$$\frac{\partial^2 G}{\partial y_i \partial y_j} = \frac{\partial (1)}{\partial y_j} = 0$$

λ so on: $(-1)^3 \cdot (0) = 0 \geq 0$

Nested logit

$$G = \left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda} + \left(\sum_{j \in NF} y_j \lambda^{\lambda} \right)^{\lambda} + y_0$$

Now

$$p_i = \frac{1}{G} \frac{\partial G}{\partial u_i}$$

$$\frac{\partial G}{\partial u_i} = \frac{\partial}{\partial u_i} \left[\left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda} + \left(\sum_{j \in NF} y_j \lambda^{\lambda} \right)^{\lambda} + y_0 \right]$$

$$\begin{aligned} &= \lambda \left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda-1} \times \lambda \left(\sum_{j \in F} y_j \lambda^{\lambda-1} \right) \frac{\partial \sum_{j \in F} y_j}{\partial u_i} \\ &= \left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda-1} \left(\sum_{j \in F} y_j \lambda^{\lambda-1} \right) y_j \frac{\partial u_i}{\partial u_i} \quad \left[\text{since } y_j = e^{u_j/\lambda} \right] \\ &= \left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda-1} \left(\sum_{j \in F} y_j \lambda^{\lambda-1+1} \right) \\ &= \left(\sum_{j \in F} y_j \lambda^{\lambda} \right)^{\lambda-1} e^{u_i/\lambda} \end{aligned}$$

$$\Rightarrow p_i = \frac{1}{G} \frac{\partial G}{\partial u_i} = \frac{1}{G} \left(e^{u_i/\lambda} \left(\sum_{j \in F} e^{u_j/\lambda} \right)^{\lambda-1} \right)$$

$$= e^{u_i/\lambda} \left(\sum_{j \in F} e^{u_j/\lambda} \right)^{\lambda-1} \quad \left[y_j = e^{u_j} \right]$$

$$\left(\sum_{j \in F} e^{u_j/\lambda} \right)^\lambda + \left(\sum_{j \in NF} e^{u_j/\lambda} \right)^\lambda + e^{u_0}$$

Conditioning:

$$G = \left(\sum_{j \in F} y_j^{1/\lambda} \right)^\lambda + \left(\sum_{j \in NF} y_j^{1/\lambda} \right)^\lambda + y_0$$

for $y_j \geq 0$ $\lambda \geq 0$ $\lambda \geq 0$ $\lambda \geq 0$

$$\Rightarrow G \geq 0 \quad \text{as } \lambda \geq 0$$

$$G(ty) = \left(\sum_{j \in F} (ty_j)^{1/\lambda} \right)^\lambda + \left(\sum_{j \in NF} (ty_j)^{1/\lambda} \right)^\lambda + (ty_0)$$

$$= \left[t^{1/\lambda} \sum_{j \in F} (y_j)^{1/\lambda} \right]^\lambda + \left[t^{1/\lambda} \sum_{j \in NF} (y_j)^{1/\lambda} \right]^\lambda + t y_0$$

$$= t G(y)$$

$\Rightarrow$ homogeneous of degree 1

limiting

let $y_i \rightarrow \infty$ belong to Nest r

$$\Rightarrow \left(\sum_{j \in F} y_j^{1/\lambda} \right)^\lambda \rightarrow \infty$$

$\Rightarrow$ Same if $y_i \rightarrow \infty$ in Nest NF

In case $y_0 \rightarrow \infty$

Then

$$G = \left(\sum_{j \in F} y_j^{1/\lambda} \right)^\lambda + \left(\sum_{j \in NF} y_j^{1/\lambda} \right)^\lambda + y_0 \rightarrow \infty$$

$$= \infty$$

Now

$$(-1)^{s+1} \frac{\partial^s G}{\partial y_1 \dots \partial y_s} \geq 0$$

for $s=1$

Take one Nest r

$$\frac{\partial}{\partial y_i} \left(\sum_{j \in F} y_j^{1/\lambda} \right)^\lambda = \lambda \left(\sum_{j \in F} y_j^{1/\lambda} \right)^{\lambda-1} \cdot \frac{1}{\lambda} y_i^{1/\lambda-1}$$

$$= \left(\sum_{j \in F} y_j^{1/\lambda} \right)^{\lambda-1} y_i^{1/\lambda-1}$$

$$\geq 0$$

for $s=2$

$$\frac{\partial^2 g}{\partial y_i \partial y_j}$$

$$= \frac{\partial \left[\left(\sum_{j \in F} y_j^{\frac{1}{\lambda}} \right)^{\lambda-1} y_i^{\frac{1}{\lambda-1}} \right]}{\partial y_j}$$

$$= \cancel{y_i}^{\frac{1}{\lambda-1}} \times \frac{(\lambda-1)}{\cancel{\lambda}} \times \left(\sum_{j \in F} y_j^{\frac{1}{\lambda}} \right)^{\lambda-2}$$

$$\times \frac{1}{\lambda} \times y_j^{\frac{1}{\lambda}-1}$$

$$= \frac{\lambda-1}{\lambda} \left(y_i^{\frac{1}{\lambda}-1} \right) \left(y_j^{\frac{1}{\lambda}-1} \right) \left(\sum_{j \in F} y_j^{\frac{1}{\lambda}} \right)^{\lambda-2}$$

Now

$$1 \geq \lambda > 0$$

$$\Rightarrow 1-1 \geq \lambda-1 > 0-1$$

$$0 \geq \lambda-1 > -1$$

$$\Rightarrow \lambda-1 \leq 0 \Rightarrow -ive$$

Now

$$(-1)^{2+1} (-ive \text{ term}) = (-ive) \times (-ive) \\ = +ive \geq 0$$

$\Rightarrow$ satisfied

$\triangleright$ we can continue for rest of s .